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Spectra III / Review of the

functor of points

Last time More on Spectra, Stable ∞ -Categories

Today The tensor product of Spectra, review of the functor of points approach to algebraic geometry

Recall.

- > The inclusions $\mathrm{Sp} \xleftarrow{P_1} \mathrm{Fun}_*(\mathrm{An}_*^{\mathrm{fin}}, \mathrm{An}) \xleftarrow{(-)^{\mathrm{red}}} \mathrm{Fun}(\mathrm{An}^{\mathrm{fin}}, \mathrm{An})$ have left adjoints
- > The functor $\Sigma^\infty \mathrm{Sp} \rightarrow \mathrm{An}_*$ has a left adjoint $\Sigma^\infty \mathrm{An}_* \rightarrow \mathrm{Sp}$ defined by $\Sigma^\infty X = P_1(X \wedge (-))$

Thm (universal property of Spectra). Let $\mathcal{C}$ be a stable ∞ -category with colimits. Then evaluation on the sphere spectrum defines an equivalence of ∞ -categories

$$\begin{array}{ccc} \mathrm{Fun}^{\mathrm{colim}}(\mathrm{Sp}, \mathcal{C}) & \xrightarrow{\sim} & \mathcal{C} \\ F & \longmapsto & F(S) \end{array}$$

Moreover, every colimit-preserving functor $\mathrm{Sp} \rightarrow \mathcal{C}$ is a left adj

The tensor product of spectra

Goal Equip $\mathcal{S}p$ with a symmetric monoidal structure $\otimes$ so that:

- (1) S is the unit
- (2) $\Sigma^\infty An_* \longrightarrow \mathcal{S}p$ is symmetric monoidal, where An_* is given the smash product operation
- (3) Generally $\otimes$ should be 'well-behaved'

To understand the last point, we use the following as motivation.

Lemma Let A be a ring. Then the tensor product of ordinary A -modules $(-) \otimes_A (-) : \text{Mod}_A^\heartsuit \times \text{Mod}_A^\heartsuit \longrightarrow \text{Mod}_A^\heartsuit$ is the unique symmetric monoidal structure such that

- (1) A is the unit.
- (2) $\otimes_A$ preserves colimits separately in each variable.

Thm. Let A be a ring. Then the derived tensor product $\otimes_A^{\mathbb{L}}$ refines to a symmetric monoidal structure on $D(A)$. Moreover, this symmetric monoidal structure is uniquely characterized by:

(1) A (concentrated in degree 0) is the unit.

(2) $\otimes_A^{\mathbb{L}}$ preserves colimits separately in each variable.

> It turns out that the obvious variant holds for spectra.

Thm. The ∞ -category Sp admits a unique symmetric monoidal structure $\otimes = \otimes_S$ satisfying:

(1) S is the unit.

(2) $\otimes$ preserves colimits separately in each variable.

Moreover, this symmetric monoidal structure also satisfies:

(3) The functor $\Sigma^\infty: (\mathrm{An}_*, \wedge) \longrightarrow (\mathrm{Sp}, \otimes)$ is symmetric monoidal.

(4) This symmetric monoidal structure is closed so there is a functor $\text{Hom}_{\text{Sp}}: \text{Sp}^{\text{op}} \times \text{Sp} \rightarrow \text{Sp}$ and a natural equiv

$$\text{Map}_{\text{Sp}}(E \otimes F, G) \simeq \text{Map}_{\text{Sp}}(E, \text{Hom}_{\text{Sp}}(F, G)).$$

Additionally, the composite

$$\text{Hom}_{\text{Sp}}: \text{Sp}^{\text{op}} \times \text{Sp} \longrightarrow \text{Sp} \xrightarrow{\Omega^\infty} \text{Sp}$$

is equivalent to Map_{Sp} .

Remark Traditionally, $\otimes$ was called the smash product of spectra and denoted by $\wedge$.

Q How can we construct this $\otimes$? The solution is to consider the more general problem of constructing symmetric monoidal structures on functor ∞ -categories.

Day Convolution

> We need a preliminary definition/axiom

Def. Let $\mathcal{C}$ and I be ∞ -categories, and assume that $\mathcal{C}$ has pullbacks and I -shaped colimits. We say that I -shaped colimits are universal in $\mathcal{C}$ if for all diagrams $X: I \rightarrow \mathcal{C}$ and maps

$$\operatorname{colim}_{i \in I} X_i \longrightarrow Z \longleftarrow Y,$$

the natural map

$$\operatorname{colim}_{i \in I} (X_i \times_Z Y) \longrightarrow \left(\operatorname{colim}_{i \in I} X_i \right) \times_Z Y$$

is an equivalence.

Axiom. All small colimits are universal in $\mathcal{A}_n$. In particular, the functor $(-) \times (-): \mathcal{A}_n \times \mathcal{A}_n \rightarrow \mathcal{A}_n$ preserves colimits separately in each variable.

Thm(Day). Let $(\mathcal{C}, \otimes_{\mathcal{C}}, 1_{\mathcal{C}})$ be a small symmetric monoidal ∞ -cat and let $(\mathcal{D}, \otimes_{\mathcal{D}}, 1_{\mathcal{D}})$ be a symmetric monoidal ∞ -category with colimits such that $\otimes_{\mathcal{D}}$ commutes with colimits separately in each variable. Then $\text{Fun}(\mathcal{C}, \mathcal{D})$ admits a natural symmetric monoidal structure $\otimes_{\text{Day}}$ called Day convolution with

(1) $1_{\text{Day}}: \mathcal{C} \rightarrow \mathcal{D}$ the left Kan extension of $1_{\mathcal{D}} * \rightarrow \mathcal{D}$ along $1_{\mathcal{C}}:$

$$\begin{array}{ccc} * & \xrightarrow{1_{\mathcal{D}}} & \mathcal{D} \\ 1_{\mathcal{C}} \downarrow & \Downarrow & \nearrow \\ \mathcal{C} & \dashrightarrow & \mathcal{D} \end{array} \quad 1_{\text{Day}}$$

$$1_{\text{Day}}(c) = \text{Colim}_{1_{\mathcal{C}} \rightarrow c} 1_{\mathcal{D}}$$

(2) For functors $F, G: \mathcal{C} \rightarrow \mathcal{D}$, the tensor product $F \otimes_{\text{Day}} G$ is the left Kan extension of $\mathcal{C} \times \mathcal{C} \xrightarrow{F \times G} \mathcal{D} \times \mathcal{D} \xrightarrow{\otimes_{\mathcal{D}}} \mathcal{D}$ along $\otimes_{\mathcal{C}}: \mathcal{C} \times \mathcal{C} \rightarrow \mathcal{C}$. Explicitly,

$$(F \otimes_{\text{Day}} G)(c) := \text{Colim}_{c' \otimes_{\mathcal{C}} c'' \rightarrow c} F(c') \otimes_{\mathcal{D}} G(c'')$$

Remark. If $(D, \otimes_D) = (An, \times)$, this symmetric monoidal structure is also closed.

Ex. If we take $(e, \otimes_e, 1_e) = (An_*^{fin}, \wedge, S^0)$ and $(D, \otimes_D, 1_D) = (An, \times, *)$, then this gives $\text{Fun}(An_*^{fin}, An)$ a closed symmetric monoidal structure $\otimes_{\text{Day}}$.

Warning. If E and F are spectra, then $E \otimes_{\text{Day}} F$ need not be a spectrum.

Thm. The ∞ -category Sp admits a unique symmetric monoidal structure that makes

$$p_1^{\text{red}}: \text{Fun}(An_*^{fin}, An) \longrightarrow \text{Sp}$$

symmetric monoidal where the source is given the sym. mon. structure of Day convolution. Moreover, this sym. mon. struct. satisfies all of the earlier stated properties.

Where we're headed

So far we have $\mathcal{S}p$ with its symmetric monoidal structure
so we can talk about commutative (bi)algebras

> We have the symmetric monoidal functor

$$\mathcal{S}\langle - \rangle = \sum_{+}^{\infty} A_n \longrightarrow \mathcal{S}p,$$

so for any commutative monoid in anima (or just Sets), we get a commutative bialgebra in Spectra

> So we can do the affine version of toric geometry over $\mathcal{S}$

> We need an analogue of scheme theory

- One approach: try to formulate a "locally spectrally ringed space" and copy how things work usually

- Better approach: avoid the underlying space, and use the functor of points

Review of the functor of points perspective

Notation. For convenience, in this part of the lecture, for a 1-category $\mathcal{C}$, we write $\text{Psh}(\mathcal{C}) := \text{Fun}(\mathcal{C}^{\text{op}}, \text{Set})$ for presheaves of Sets.

Recall. For a scheme X , the functor of points of X is just the presheaf $X(-) = \mathcal{L}(X)$ on Sch represented by X .

> Since the Yoneda embedding is fully faithful, $X(-)$ determines X .

Q. How can we isolate Sch inside of $\text{Psh}(\text{Sch})$ in a geometric way?

> First, every scheme X satisfies Zariski descent.

Def. A presheaf $F: \text{Sch}^{\text{op}} \rightarrow \text{Set}$ is a Zariski sheaf if for every scheme X and open cover $(U_i)_{i \in I}$ of X , the natural diagram

$$F(X) \longrightarrow \prod_{i \in I} F(U_i) \rightrightarrows \prod_{i, j \in I} F(U_i \cap U_j)$$

is an equalizer

- > We write $\text{Sh}_{\text{zar}}(\text{Sch}) \subset \text{PSh}(\text{Sch})$ for the full subcategory spanned by the Zariski sheaves
- > Since every scheme admits a cover by affines, one expects Zariski sheaves to be determined by their values on affines. Indeed, this is true

Def A functor $F: \text{CAlg} \simeq \text{Aff}^{\text{op}} \rightarrow \text{Set}$ is a Zariski sheaf if for every ring A and finite set of elements $(f_i)_{i \in I}$ that generate A , the natural diagram

$$F(A) \longrightarrow \prod_{i \in I} F(A_{f_i}) \rightrightarrows \prod_{i, j \in I} F(A_{f_i f_j})$$

is an equalizer

- > This is just descent for the open cover $(D(f_i))_{i \in I}$.
- > We write $\text{Fun}_{\text{zar}}(\text{CAlg}, \text{Set})$ for the full subcategory of Zariski sheaves

> The following is a special case of the 'comparison Lemma' for sites

Lem Restriction along $\text{Spec CA} \longleftrightarrow \text{Sch}^{\text{op}}$ defines an equiv. of categories

$$\text{Sh}_{\text{zar}}(\text{Sch}) \xrightarrow{\sim} \text{Fun}_{\text{zar}}(\text{CA}, \text{Set})$$

The inverse sends a zariski sheaf $F: \text{CA} \rightarrow \text{Set}$ to its right kan extension along $\text{Spec CA} \longleftrightarrow \text{Sch}^{\text{op}}$

$$X \longmapsto \lim_{\text{Spec}(A) \rightarrow X} F(A)$$

> We need to characterize admitting an open cover by affines

- we start by describing covering intrinsically to zariski sheaves

Lem. Let X be a scheme and $(U_i)_{i \in I}$ a collection of open subschemes of X . TFAE:

(1) $(U_i)_{i \in I}$ is an open cover of X .

(2) For each local ring A , we have $X(A) = \bigcup_{i \in I} U_i(A)$.

(3) For each field k , we have $X(k) = \bigcup_{i \in I} U_i(k)$.

(4) The induced map $\bigsqcup_{i \in I} U_i(-) \rightarrow X(-)$ is an epimorphism in $\mathbf{Sh}_{\text{zar}}(\mathbf{Sch})$.

Note we only need $(1) \Leftrightarrow (4)$, but (2) and (3) are also useful facts.

> We need to understand openness.

a ring object in $\mathbf{Fun}(\mathbf{CAlg}, \mathbf{Set})$

Def (functions). Write $A^1: \mathbf{CAlg} \rightarrow \mathbf{Set}$ for the forgetful functor. For a functor $F: \mathbf{CAlg} \rightarrow \mathbf{Set}$, the ring of functions on F is the ring

$$\mathcal{O}(F) = \text{Map}(F, A^1).$$

Notation. For a ring A , let's somewhat abusively write $\text{Spec}(A)$ for the functor of points

$$\text{Spec}(A)(-) = \text{Map}_{\text{CAlg}}(A, -) : \text{CAlg} \rightarrow \text{Set}$$

Def (nonvanishing locus). Let $F : \text{CAlg} \rightarrow \text{Set}$ be a functor and $S \subset \mathcal{O}(F)$ a set of functions. The nonvanishing locus of S is the subfunctor $D(S) \subset F$ defined by

$$D(S)(A) := \{x \in F(A) \mid (f(x))_{f \in S} \text{ generates } A\}$$

$$\begin{array}{ccc} f: F & \longrightarrow & A^1 \\ & & \downarrow \\ & & F(A) \longrightarrow A \\ & & \downarrow x \longmapsto f(x) \end{array}$$

Exercise. If $F = \text{Spec}(A)$ and $S \subset A$ is a collection of elements, check that $D(S)$ is the functor of points of the open subscheme typically denoted $D(S) \subset \text{Spec}(A)$

Def (open subfunctor). Let $F : \text{CAlg} \rightarrow \text{Set}$ be a functor and $U \hookrightarrow F$ a subfunctor. We say that U is an open subfunctor if for every map $x : \text{Spec}(A) \rightarrow F$, the pullback

$$\begin{array}{ccc}
 x^{-1}(U) & \longrightarrow & U \\
 \downarrow \perp & & \downarrow \\
 \text{Spec}(A) & \xrightarrow{x} & F
 \end{array}$$

is isomorphic to $D(S)$ for some $S \subset A$.

Thm. The functor

$$\begin{array}{ccc}
 \text{Sch} & \longrightarrow & \text{Fun}_{\text{zar}}(\text{CAlg}, \text{Set}) \\
 X & \longmapsto & X(-)
 \end{array}$$

is fully faithful with image those Zariski sheaves F such that there exists a collection of open subfunctors $(U_i)_{i \in I}$ such that:

- (1) Each U_i is representable, i.e., $U_i(A) \cong \text{Map}_{\text{CAlg}}(B_i, A)$ for some ring B_i .
- (2) The induced map $\bigsqcup_{i \in I} U_i \longrightarrow F$ is an epimorphism in $\text{Fun}_{\text{zar}}(\text{CAlg}, \text{Set})$.